Carleton University

COMP4905 – Honours Project

Parked domain name detection based on DNS and network characteristics

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Abstract

The aim of the project is to identify parked domains using DNS and network-related characteristics. A parked domain is defined as a domain that has been registered for a long period of time (usually at least a few months) without hosting any active web content or online services, and is usually reserved for future resale at a high price. In this study, I focused on analysing the current situation of wild parked domains. To do this, I first downloaded and processed a large DNS dataset locally from the OpenIntel platform. A Python program was then developed to determine the registration status of the domain names using the python-whois library and to make a preliminary classification based on whether the registration had been in place for more than three months. I then used dnspython to build a DNS query tool to further identify potential parked domain names by analysing DNS records. The results showed that about 5% of the domain names tested met my definition of parked domain names.

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Chapter 1

Introduction

As the domain name market continues to grow, a large number of parked domains have appeared on the Internet. A parked domain is a domain name for which the registrant does not plan to host any actual web content or online services for an extended period of time (usually at least a few months). Even if a domain name only points to a blank page, it is still considered a parked domain. Usually, the domain name holder intends to resell the domain name at a later date for a higher price.
Traditional research on parked domains has tended to focus on domains that point to typical parked services or parked web pages, such as those that redirect to third-party advertisement pages or sites that explicitly indicate that the domain is for sale. However, the scope of parked domains is not limited to this: many domains do not link directly to any of the common parked services, but remain in a state of perpetual inactivity with no actual content or services, or the domain is not being used for the purpose of displaying web pages.

In the current domain name ecosystem, broadly parked domain names have the following characteristics:

- **Long-standing registrations with no real content:** The domain name owner may be holding the domain name for hoarding or resale purposes, without building any websites or services.

- **Lack of DNS configuration information:** Except for the NS record, the domain name has almost no A record, MX record, CNAME record, etc., which can be regarded as an "empty shell" domain name.

- **Missing or undisclosed information:** Sometimes we are unable to find the registration information of some domain names. These domains for which no information can be found are not necessarily parked domains. Some domains may not be publicly available due to policy issues.

Compared with studies that only examine known parking services or advertising pages,

this project looks at parking domains from a broader perspective. It uses WHOIS data and DNS record characteristics to automatically identify and classify a large number of domain names. This research can offer a more comprehensive basis for future work in domain name ecology, network security, and resource management.

Chapter 2

Related Work

In today's increasingly complex network environment, many domain names that were originally idle or used only for placeholders have gradually evolved into parked domain names. These parked domains generate revenue from traffic through advertising or simple redirection. These domains differ significantly from regular websites in terms of page display, resolution configuration, and actual usage. The detection methods for parked domains can be roughly divided into three categories: identification based on web content, identification based on DNS data, and identification using multi-feature analysis. The following will introduce and discuss the work in each of these categories.

2.1 Web Content Based Identification of Parked Domain

2.1.1 Penetration Analysis and URL Pattern Stamping

This study [11] defines domain parking as a model that makes money from advertising. This means that the owner of the domain places advertisements on the domain page without actually building a website, and earns revenue through user clicks or redirects.
To gain insight into how such parked domains work, the researchers used dynamic crawlers and reverse DNS lookups to collect over 100,000 possible parked domains and simulated 24 million visits over a five-month period. The core detection method is infiltration analysis. First, access traffic was initiated in a controlled environment, such as a virtual machine, and all hops related to parked domains were recorded. Next, a modified browser is used to capture advertised pages, HTTP requests and bounce information. Traffic is then purchased or clicks are introduced and observed to see if they actually generate advertising revenue to confirm how the parked domain is profiting. Finally, we cluster a large number of captured URLs/IPs to extract similar advertising chain patterns and identify domains that may belong to the same ad network or parking system.

Through in-depth analysis of this data, the researchers proposed URL Pattern Stamping. This method involves extracting important parameters and keywords from redirect links. If multiple redirect chains are found to present the same pattern, it means that they may belong to the same advertiser or have the same monetisation method.
The study found that a large amount of traffic generated by Pay-Per-Redirect is actually traffic spam, accounting for about 70.7%. In addition, 5% of visits were linked to malicious activities. However, the researchers also pointed out that since they only analysed a small number of advertising networks, and due to possible situations such as crawlers being identified or redirection chains being hidden, the actual violation rate may be higher. Overall, the study successfully revealed a number of illegal traffic issues in parked domains through a combination of penetration analysis, dynamic crawler, and URL pattern stamping. There are many types of advertising networks and the parking service is constantly evolving. This means that a larger-scale, more flexible detection method is still needed to keep up with the complete coverage of all types of domain parking.

2.1.2 ML-Base Detection

This study [8] defines parked domains as domains that have not yet launched actual content and only place advertising links or banners for revenue. The researchers proposed a method that uses multiple data features and a machine learning model to detect them. The researchers collected information on multiple dimensions based on more than 8 million suspected parked domains, including DNS resolution records, WHOIS information, HTML web page source code, HTTP request traces (HAR files), and redirect chain data. They also further annotated whether there were misspelled domains.
During the core detection process, the researchers selected 3,000 parked domain names and 3,000 non-parked domain names as training samples. They extracted 21 features, input them into a random forest classifier, and used 10-fold cross-validation to evaluate the model performance. The final results showed that the detection accuracy was as high as 98.7%, and the false alarm rate was only 0.5%, indicating that these features can be used to quickly and accurately distinguish parked domains containing only advertisements.
However, the researchers also point out some potential limitations: for example, the training set may be over-fitted to a specific parking pattern. This means that when new parking methods emerge, the model's effectiveness may decline. In addition, although the current false positive rate is very low, it still needs to be continuously verified and optimised in different environments and larger-scale deployments.

2.1.3 Detection Based On Multilingual Keyword Extraction

The study [6] defines parked domains as domains that have not been put back into use after being bought, cancelled, suspended or expired, and whose web pages no longer dis-

play the original content but only advertisements.
The researchers first selected 165 domains from the Dot.tk database that had been confirmed to be parked. For each domain name, they collected the previous page summary, anchor text, source link, and information such as the webpage body and title obtained through a crawler. Then, they used NLP (natural language processing) tools to preprocess the multilingual text, such as word segmentation, stopword filtering, and part-of-speech tagging, and combined techniques such as TF-IDF and BM25F to measure the keyword value of different positions on the webpage.
On this basis, the researchers used an adaptive algorithm based on Multi-Armed Bandits. The researchers adjusted and optimised the initially extracted keywords in real time. This allowed the system to continuously fine-tune the keyword list based on user clicks or other feedback, with the aim of improving ad click-through rates or page relevance. To verify the effectiveness of these algorithms, they also invited 23 evaluators to subjectively score the match between keywords and the actual content of the page on a crowdsourcing platform, and calculated the overall accuracy and consistency coefficient.
The results show that this strategy of combining BM25F with a multi-armed bandit not only has higher keyword extraction accuracy, but also better copes with the limitations of low-quality tags or incomplete page text that are common in parked domains. However, the study also highlighted real-world challenges. For example, many HTML meta tags in parked domains are not reliable, word segmentation and semantic understanding in non-English environments remain difficult, and crowdsourced evaluation can introduce some subjective bias. The keyword model itself also needs to be updated regularly to adapt to changes in user interests and web page layouts.

2.1.4 Comparison of research methods

The three studies have the following in common:

- All three papers define parked domains as domains that do not host real website content and only generate revenue through advertising or redirects.
- The studies have all sampled and analysed parked domains at large scale. Including domains ranging from thousands to millions and beyond, and have collected relevant features through active crawling or simulated visits.
- All three papers discuss limitations such as parking domain hiding mechanisms, multilingual environments, and bot identification. The collected data may not be complete, or the model may be limited by the current feature engineering and training set distribution.

These three studies differ in the following ways:

- Research [11] focuses more on revealing the traffic fraud and advertising chain profit model behind parked domains.
- The main objective of research [8] is to quickly identify parked domains using a highly accurate classifier.
- Research [6] focuses on how to improve the matching between keywords and user clicks, thereby increasing the profitability of parked domains.
- The three studies used different technical routes.

2.2 DNS Based Identification of Parked Domain

2.2.1 DNS Clustering

This study [7] focuses on detecting parked domains by analysing the relationship between domain names and authoritative DNS servers without relying on crawling web content. The authors define parked domains as an automated web page generation mechanism. This mechanism refers to a specific parked service provider that provides authoritative DNS servers and resolves domain names to an IP address range associated with advertising.

To complete the study, the researchers collected three types of core data. DNS server information is used to discover which domain names are collectively pointing to the same or similar parked service DNS. IP address data is used to observe the concentration of resolution results, such as a large number of domain names sharing the same IP. Domain name data and domain name semantics and spelling variants provide auxiliary features for subsequent clustering.

To determine which domains are parked, the authors use a semi-supervised clustering approach. The semi-supervised clustering approach first selects suspected parked domains based on semantics, spelling, or parsing results. Next, the researchers cluster the authoritative DNS servers to which these suspicious domains point to identify typical parking service providers. For any domain name that resolves to a DNS server marked as a parking service, the researchers then confirm whether it is parked based on other characteristics.

The experimental results show that this method can be used to identify parked domains with an accuracy of about 92.8% without analysing the content of the webpage. The identification process is faster and suitable for large-scale monitoring. This detection method is fast and highly real-time, and is effective for most common parking service providers. However, when the parking service provider is more niche or significantly different from

the regular service provider, it may not be successfully identified. In addition, this detection method currently lacks experimental evaluation in different regions, different sizes, and multi-language environments.

2.2.2 A Large-Scale DNS-Based Detection

In this study [5], the authors define parked domains as a type of domain name usage that is primarily aimed at advertising and traffic monetisation. Specifically, these domains do not usually provide actual content or services for users, but instead display automatically generated advertising pages or placeholder pages where normal web pages should be displayed. Compared to ordinary websites with original content and complete functions, parked domains focus more on how to make revenue from the remaining traffic, so the page structure is often relatively simple or even highly repetitive.

In order to systematically quantify the size and distribution of parked domains worldwide, the researchers used large-scale DNS data analysis methods. The data included DNS records, large-scale DNS scan data, and public suffix lists (PSLs). The researchers collected DNS records to determine whether they pointed to the name servers or dedicated IP segments of a specific parking service provider. For large-scale DNS scan data, the researchers used the TUM, OpenINTEL, and Rapid7 datasets. The researchers believe that cross-referencing data from different sources can help improve identification accuracy. Public suffix lists can be used to help distinguish between different top-level domains and deeper domain structures, which facilitates subsequent statistical analysis of which types of suffixes have more parked domains.

In terms of the core detection method, researchers first check whether the NS record or IP address of a domain name matches those of a known parking service provider. Then, they combine multi-source scan data, including test data from tools such as Common Crawl, to perform a more comprehensive verification. Through this large-scale cross-analysis, the study ultimately identified about 60 million parked domains.

However, there are some limitations to the research. First, for parking services with unconventional or mixed configurations, the DNS record-based method may miss some. Second, the scanning cycles and methods of different data sources are not consistent, which may lead to inaccurate understanding of the time-series changes of some domain names.

2.2.3 Detection Based on Private IP Resolution and Timing Analysis

In this study [10], the authors define 'parked domains' as private IP addresses that resolve to a nonsense location when the domain is not actually in use, to hold the space and keep

it ready for online use at any time.
To study this parking of domain names, the research team used large-scale passive DNS data. This passive DNS data covers 610 million unique fully qualified domain names, and the research focused on the resolution records of private IP address segments.
In terms of detection methods, they first filter out all non-A records, and then time-sequence trace the remaining data. If a domain name is found to repeatedly switch between private and public IP addresses at different times during the tracking process, it is judged to be in a parked state. At the same time, the researchers also refer to a blacklist of malicious domain names. The blacklist compares the detected suspicious domain names to confirm whether they pose a potential security threat. Domain names that do not show obvious signs of malicious intent but match the resolution pattern are manually reviewed and added to the whitelist to reduce the false positive rate.
The results show that about 0.0035% of the domain names in the dataset are confirmed as parked domain names, and 7.3% of them appear in the blacklist. This shows that the vast majority of parked domain names do not directly exhibit malicious behaviour.
There are some limitations to this research. First, this method relies on manual review and blacklist comparison, which is not cost-effective. Second, long-term observation is required to capture the temporal changes in IP resolution, and real-time detection is difficult. Third, the coverage of the blacklist is limited, and if it is not updated in time, it will also affect the accuracy.
This study provides preliminary evidence that it is indeed possible to capture some parked domains using private IP address ranges and time series information.

2.2.4 Comparison of research methods

The three studies have the following in common:

- All three studies identify parked domains by analysing domain name DNS resolution records, authoritative DNS server information, IP allocation and other network-layer characteristics.
- All three studies involve a large amount of DNS data from multiple sources and extract the characteristics of parked domains or measure the distribution of parked domains.
- They all use the concentration of a large number of domain names pointing to the same batch of DNS service providers or specific IP segments to determine whether they are operated by the same parking service provider.
- The identification effect on small or mixed parking service providers is reduced.

- For domains that are not yet in the known database, there is a risk of underreporting or misjudgment.
- Further experiments or verification are required in terms of different regions, different language environments, and data timeliness.

These three studies differ in the following ways:

- Research [7] identifies typical parking service providers by clustering DNS servers for suspicious domain names, and then marks the domain names resolved to these service providers as parked.
- Research [5] collects and fuses multiple DNS data to find domain names that point to the NS or IP of known parking service providers, and cross-verifies them with publicly crawled data.
- Research [10] determines the parking status by analysing whether the domain name is repeatedly switched between private IP and public IP at different times, and combines it with blacklists and manual reviews.

2.3 Multi-feature analysis

2.3.1 Detection Based on Multi-source Data and Time Series Analysis

In this study [9], the authors define domain parking as a domain usage method that places advertisements on otherwise unused domains through domain parking services.
The researchers collected DNS forward resolution data. The data contains 160 million unique FQDNs and 210 million e2LDs to provide the basis for parking detection. Next, the researchers collected WHOIS data. Multiple snapshots were used to track changes in domain name registration information and identify the status of the domain name at different stages. The researchers then compared the domain name with commercial or public blacklists to exclude Alexa Top sites to reduce the false positive rate. A domain is initially determined to be parked if the NS record points to the authoritative DNS of a known parking service provider. If the domain appears in a blacklist and is not among the Alexa Top sites, there may be a risk of malicious activity. Finally, a time series analysis was performed on these parked domains. It included examining the track of domain names from parking to malicious, or from malicious to parking, and the behavioural patterns of switching back and forth between multiple parking service providers.
The study found that there is indeed a correlation between parked domains and malicious domains. The study found that 3,964 domains were converted into malicious domains

after using the parking service. There were a total of 107,351 records of the conversion of parked domains to malicious domains. This indicates that some parked domains are indeed associated with malicious activities. The study also found that 3.02 million domains have repeatedly switched between multiple parking service providers. This indicates that some domain name holders may be trying to find a better monetisation model or a more flexible traffic distribution strategy on different service platforms.
This study also has some limitations. If multiple users share the same e2LD and only the registration information at the second-level domain name level is available, it is difficult to accurately distinguish the actual usage of specific sub-domains. In the event that NS records or blacklist updates are not timely, the study can only use approximate inference, which leads to certain errors. The scanning cycle and format of different data sources differ, making it difficult to ensure that the resolution and registration information of all domain names is complete.

2.4 Advantages and disadvantages of each method

2.4.1 Base on Web Content

Advantages

- By inspecting the content of the webpage itself, it is possible to visually determine whether the page is a shell, an advertisement, or a redirection chain.
- For any type of website, the crawler can obtain web pages and perform content analysis, which is suitable for identifying different types of parking services.
- It can extract detailed information such as page structure, advertising links, redirection behaviour, and scripts, which provides a basis for subsequent malicious behaviour or security risk assessment.

Disadvantages

- Large-scale web crawling and complex text or redirection analysis are required, especially when faced with millions or even billions of domain names, which is extremely costly and computationally intensive.
- Web content may change frequently, requiring frequent repeated collection, otherwise it is difficult to maintain the timeliness of detection.
- Not all domain names provide web services. Web content detection is not effective for these domain names.

2.4.2 DNS-based analysis

Advantages

- Only DNS records for domain names need to be queried or analysed, without loading the content of the web pages. The relatively low computing and network consumption is suitable for quickly scanning large databases of domain names.
- Passive or active scanning data at the DNS level is usually regularly published by multiple public projects, which facilitates batch processing.
- Many parked domain names will point to specific DNS name servers or share an IP address. Clustering and comparison can quickly locate common parking service providers.

Disadvantages

- When only the DNS level is analysed, it is impossible to accurately distinguish between different situations, and there is a risk of missed or misdiagnosed cases.
- If the parking service provider uses a hybrid CDN or custom DNS configuration, traditional DNS pattern matching methods may become invalid.
- When new parking service providers or resolution strategies emerge, methods based on DNS fingerprints or fixed IP segments require regular maintenance and updates.

2.4.3 Multi-feature analysis

Advantages

- Combining multiple dimensional features of domain names can effectively reduce misjudgments caused by a single source of information.
- Different domain name features can be used to complement and verify each other.
- It can extract detailed information such as page structure, advertising links, redirection behaviour, and scripts, which provides a basis for subsequent malicious behaviour or security risk assessment.

Disadvantages

- DNS scanning, bulk WHOIS queries, and cross-source data management need to be established. The data formats of data from different sources are prone to missing or conflicting.
- A large amount of data needs to be stored locally or in the cloud for analysis.

2.5 Rationale for the Research Method Selection

Previous literature proposed a variety of detection and identification schemes. These schemes include methods based on web content analysis, DNS record analysis, and multi-source feature analysis. After studying the existing literature and the current state of research, I finally chose a research scheme that combines DNS data with domain name WHOIS information. I chose this scheme because it can detect parked domains on a large scale with preliminary, high efficiency, and a certain degree of accuracy at an acceptable computational and data cost.

First of all, from the perspective of data acquisition difficulty, a DNS-based detection solution is more feasible in practice. Currently, there are multiple public or commercial platforms that regularly collect and publish large-scale DNS datasets, such as OpenIntel, which I used this time. These platforms usually cover mainstream top-level domains. Compared with crawling web content one by one, obtaining DNS data is significantly more efficient. Although the number of DNS records is huge, their structure is simpler, which facilitates batch storage and processing. And DNS-level detection can use existing scans, avoiding the need to invest a lot of resources at the beginning of research to build a collection system from scratch. Since this research needs to screen tens of thousands of domain name lists, the availability and ease of processing of the DNS data source is particularly critical.

Second, although crawlers and web content analysis can more accurately identify advertisements, redirect chains and positioning mechanisms on a page, the cost and risk are relatively high. Identification of parked domains based on web content often requires dynamic crawlers, multiple visits, and analysis of the text and links on the response page. This involves not only considering the huge network bandwidth consumption when faced with a large number of domain names, but also dealing with potential blocking or security mechanism detection. To evade security detection mechanisms, I may need to repeatedly access the same domain name at different times and from different IP addresses, which greatly increases the complexity and resource investment of the research. In contrast, a strategy based on DNS data does not require actual 'access to web content' in the initial stage. The research only needs to perform regular DNS resolution and registration information queries. Although this method may not be able to capture extreme disguises or obfuscations, for most parking services, there are usually obvious characteristics at the DNS level, which is sufficient for the purpose of large-scale screening.

Third, the combination of WHOIS and DNS information can effectively filter out domains that have not yet been put into use or are only being used for a short period of time. Parked domains often have the characteristic of not actually providing content for a long period of time. Therefore, this study introduces WHOIS queries in addition to DNS data.

On the one hand, it can confirm whether the domain name has actually been registered. On the other hand, it can also be used to determine whether the domain name has been continuously owned for a relatively long period of time based on the registration time. If a domain name has only been registered or renewed in the past few weeks, it cannot be determined that it is in a parked state. If it has been registered for more than three months but there is no record of any real service, it is most likely in a parked or idle state for trading purposes. This dual judgement based on duration and lack of DNS resolution can effectively reduce the misjudgement of domains that are being configured.

Fourth, ideally, we can integrate crawler content, DNS history, registration information, and other information (such as blacklist databases), and then use machine learning models to improve recognition accuracy. Such multi-feature fusion schemes do indeed lead to more refined detection results in papers and practical applications. However, in the initial stages of large-scale research, the effort required to collect and clean multi-source data is very high, and data storage and processing systems require higher-specification hardware and software configurations. At the same time, the update cycles and collection methods of different data sources are not uniform. Therefore, this study prefers to start with the relatively stable DNS and WHOIS data that are easy to obtain in the early stage, and first conduct a broad and preliminary detection and statistics. If conditions permit in the future, the web content and redirection chain information of domain names suspected of high risk or suspicious behaviour can be collected in a more focused manner to conduct more in-depth behaviour analysis and tagging.

Finally, from the perspective of extended and sustainable research, this research plan is also more flexible. DNS resolution results and domain name registration information are all basic data related to domain names, which are timely and structured. After completing the large-scale identification of suspected parked domain names, a variety of follow-up analyses can be performed. For example, port detection and web content analysis. In contrast, if multi-source in-depth detection is required from the beginning, it may require excessive resources in the early stages and is difficult to adapt to research and verification at different stages.

In summary, this study chose to use a solution that combines DNS information with bulk WHOIS queries, mainly based on the following considerations:

- Take full advantage of the availability and high coverage of large-scale DNS data to facilitate the rapid screening of massive amounts of domain names.
- Combine WHOIS registration duration to exclude domains that are still in the short-term registration cycle, and focus more on targets that meet the characteristics of parking.
- Complex web crawling processes and redirection analysis are avoided, saving research resources and reducing the risk of being blocked or masked by security

mechanisms

- There is room for further analysis, and more detailed detection of specific parked domain names can be carried out on this basis.

Chapter 3

Methodology

The objective of this study is to identify parked domain names from a broad perspective. Accordingly, the methodology is divided into three main stages: data acquisition and preprocessing, filtering of domain registration information, and DNS record querying and analysis. In this process, data from OpenIntel [1] is required as well as batch querying and classification conducted locally.

3.1 Data Acquisition and Preprocessing

Data Sources

This project first obtained DNS data for October 2023 from OpenIntel. OpenIntel regularly collects DNS records for major top-level domains (TLDs) worldwide and publishes them in an analyzable format.

File Format and Reading

OpenIntel uses specialized file formats to store massive DNS measurement results. With appropriate data I/O tools, these raw files can be decompressed and read line by line, extracting domain names and their corresponding records. When reading the data locally, core fields are extracted according to research needs and stored in a more convenient data structure for subsequent batch processing and querying.

Preliminary Filtering and Deduplication

The original data contained multiple records for the same domains, so it was necessary to de-duplicate the domains to obtain a relatively unique set of domains. Given the size of the study and the limitations of computational resources, a random sample of domains will be selected for subsequent analyses.

3.2 Filtering of Domain Registration Information

Motivation for Collecting Domain Registration Information (WHOIS)

An important prerequisite for determining whether a domain is parked is that the domain has been registered yet remains without actual content for an extended period. If a domain is not registered, the concept of "parking" does not apply. Therefore, it is necessary to retrieve the domain's "registration time" or "last update time" and compare it with the current time to confirm whether the domain has been registered for a certain duration (typically set to three months or longer).

Batch Querying and Result Parsing

In order to handle a large number of domain names, an automated method is used to query the registration information for each domain name in order. The query process uses the built-in WHOIS server corresponding to each TLD included in the python-whois [2] module to retrieve the response. The raw text obtained from these queries typically consists of several fields such as domain name, registrar, creation time, renewal time, and expiration time.

Classification Rules and Exception Handling

After obtaining the registration information, each domain is evaluated as follows:

- **Not registered or unqueryable:** If no relevant information is found in WHOIS or if the information is empty, the domain is considered not registered or unavailable.
- **Registered for more than three months:** If the creation time or the last update time is at least 90 days ago, the domain is marked as "registered and older than three months".
- **Registered for less than three months:** If the update (or creation) time is less than 90 days old, the domain is temporarily considered "registered but not yet three months old".
- **Unclear information:** A domain name that is subject to a whois search and an error occurs during the search is categorised as "Unclear information".

After each batch of domains is queried, the results are summarized and written to a structured file to facilitate subsequent analysis and effectively prevent duplicate queries.

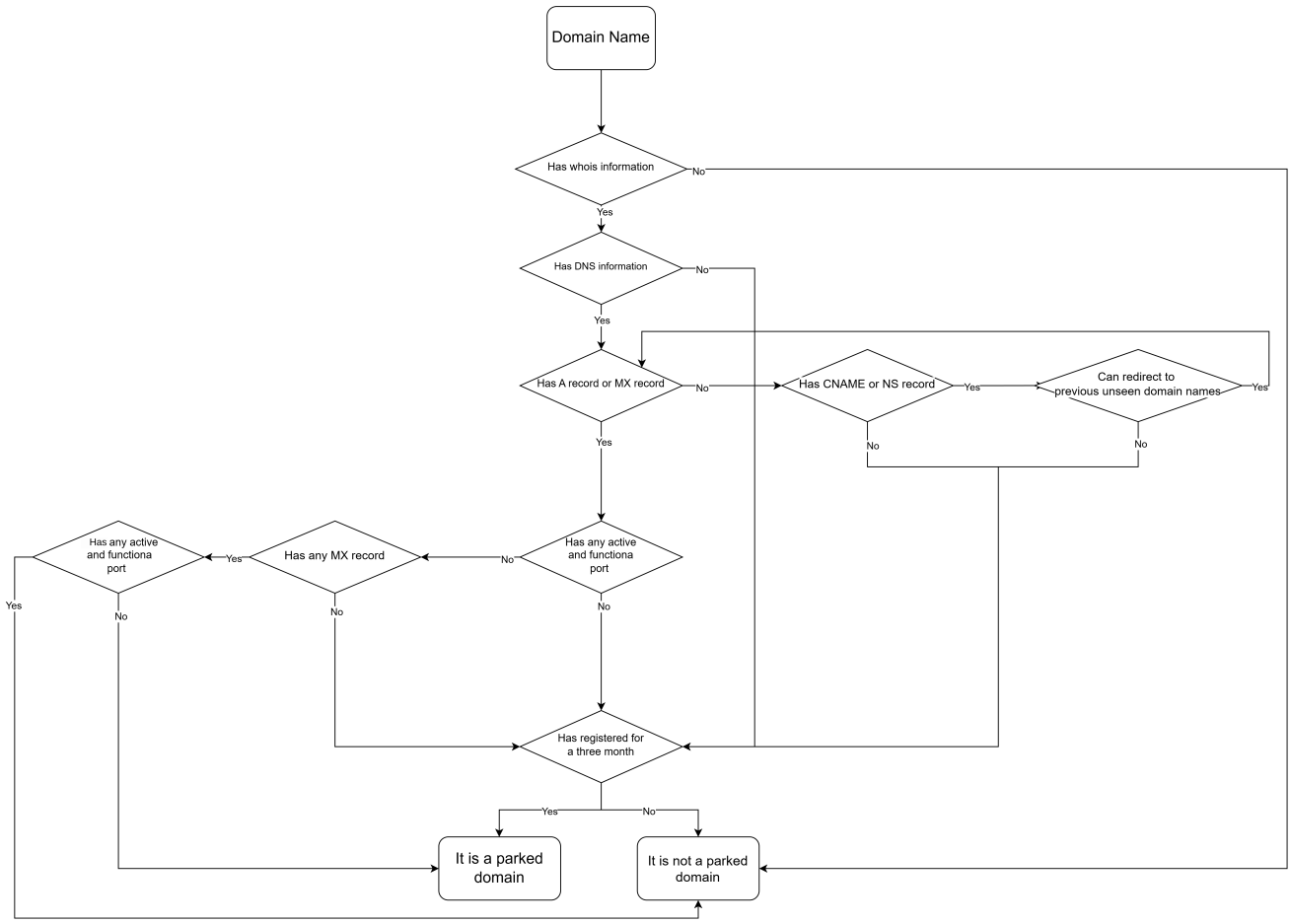

Figure 3.1: Flowchart of the algorithm for determining parked domain names.

3.3 DNS Record Querying and Analysis

After the preliminary filtering of domain registration information, for domains that have been registered for more than three months, further investigation of their DNS records is required to determine whether the domain is indeed in a parked state with no actual content. To this end, the study adopts a multi-step, multi-layer DNS record resolution strategy.

The DNS search in this study is implemented using python-dns [3]. This module builds DNS query messages in accordance with the DNS protocol standard and automatically reads a list of DNS servers configured in the system. By default, the resolution request is sent to the DNS server via UDP, and if the response is truncated due to an excessive amount of data, the request is automatically resent using TCP. After receiving the response, the module parses the binary message and extracts the required DNS information.

Step-by-Step Querying Strategy

```
try:
    answer = resolver.resolve(domain, 'A')
...
try:
    answer = resolver.resolve(domain, 'NS')
```

- **A Record:** First, attempt to resolve the domain's A record. If a valid IPv4 address is obtained, it indicates that the domain may have a basic network service entry.
- **MX Record:** If the domain has no A record, or if the A record is empty, attempt to retrieve its MX record. Valid mail server information indicates that the domain is at least used for email communication, and it can be considered to have practical use.
- **CNAME Record:** If neither of the above yields a result, query the CNAME record to determine whether there is an alias or redirection. If the CNAME record successfully points to a new domain, continue resolving the new domain until a valid record is obtained or a query depth limit is reached.
- **NS Record:** If all the above queries fail, query the domain's NS record and attempt to further query for A, MX, etc., from these servers. If no valid information is obtained, the domain is considered to have virtually no usable DNS configuration.

Recursive Processing and Loop Detection

```
answer = resolver.resolve(domain, 'CNAME')
for rdata in answer:
    cname_target = str(rdata.target).rstrip('.')
    result = _resolve(cname_target, depth + 1)
```

When processing CNAME records, multiple redirections or cyclic pointers may occur. To avoid endless queries, the program sets a maximum recursion depth and maintains a list of "visited domains". Once repeated pointers or the maximum level is reached, the resolution stops.

Exception and Delay Control

```
delay = random.uniform(1, 3)
time.sleep(delay)
```

DNS queries are prone to network delays, server timeouts, or service denials. To improve query success rates and prevent being flagged as malicious scanning, appropriate timeouts are set for each query, with delays of a few seconds added if necessary. Instances where queries return empty results or error codes are logged for later analysis and debugging.

Determining Parked Domains

```
if result is None:
    output_data["No_Result_Parked.txt"]
elif "NS" in result:
    output_data["Only_NS_records_Parked.txt"]
else:
    output_data["Faild_Parked.txt"]
```

Based on the registration term criteria and actual DNS record lookups, a domain name is considered to be a "potentially parked domain name"if it has been registered for more than three months and still does not have a usable A, MX or CNAME record. Conversely, if a valid A, MX or other service record is obtained, the domain name proceeds to the next round of algorithmic testing to determine if it is a parked domain name.

3.4 Overall Workflow and Output

In summary, the methodology of this study relies on the following overall workflow:

1. **Data Acquisition and Sample Selection:** Download raw data from OpenIntel, filter and extract a unique list of domain names, and select analysis samples as needed.

2. **WHOIS Registration Information Filtering:** Categorize domains into "Not registered/Unqueryable" , "Registered $<$ 3 months", and "Registered $\geq$ 3 months", and record relevant metadata.

3. **DNS Analysis:** For domains in the "Registered $\geq$ 3 months" category, perform multi-layer DNS record queries to further distinguish domains with no actual resolution records.

4. **Parked Domain Determination:** Combine registration duration, absence of DNS records, and actual usage factors to make a preliminary identification of broadly parked domains.

5. **Statistics and Output:** Finally, compile and output all domain information and classification results, which will be further analyzed and discussed in the results section.

Chapter 4

Results

This study used 10,000 unique domain names extracted from data collected by OpenIntel on October 1, 2023 for the experiments. Using the WHOIS filtering and DNS analysis methods described above, a preliminary validation of the methodology for determining parked domain names was conducted. The following sections detail the main experimental results.

4.1 WHOIS Query and Registration Information Classification

Data Scale and Sampling

After deduplication and initial cleansing of the raw data, a random set of 10,000 valid domain names was obtained.

4.1.1 Registration Status Determination

For the 10,000 domains, batch WHOIS queries were performed to retrieve the creation or last update times, which allowed categorization into the following three groups:

Category	Count	Percentage
Not yet registered or no WHOIS information found	2,471	24%
Registered for less than three months	1,873	19%
Registered for more than three months	5,656	57%
Total	**10,000**	**100%**

Table 4.1: WHOIS-based classification of sampled domains

- **Not registered or unqueryable:** 24% (2,471 domains).

- **Registered for less than three months:** 19% (1,873 domains).
- **Registered for more than three months:** 57% (5,656 domains).

4.1.2 Impact of ccTLDs

Among the 2,471 domains that were not registered or whose information could not be obtained, 60% originated from ccTLDs (country-code top-level domains). This indicates that some ccTLDs do not publicly disclose WHOIS information or that their registration details are difficult to retrieve via public queries.

Category	Count	Percentage
ccTLD	1,497	60%
generic top-level domain	974	40%
Total	**2,471**	**100%**

Table 4.2: Breakdown of ccTLD vs. gTLD among the 105 unregistered/unqueryable domains

In summary, within the randomly selected sample, about 57% of the domains had been registered for more than three months, approximately 19% were registered for less than three months, and the remaining 24% could not be confirmed or were unregistered.

To verify whether the domains are truly in a parked state , the focus was on the 5,656 domains registered for more than three months. For these domains, multi-layer DNS queries (A, MX, CNAME, NS, etc.) were performed.

4.1.3 Query Results Statistics

Category	Count	Percentage
No DNS information	169	3%
Only NS records and no A records could be found	129	2%
Contains A records	5,226	93%
Contains MX records	132	2%
Total	**5,656**	**100%**

Table 4.3: DNS-based classification of sampled domains

- **No DNS information:** 3% (169 domains) returned no valid DNS records (i.e., A, MX, CNAME, or NS).

- **Only NS records without further resolution:** 2% (129 domains) yielded NS records, but further attempts to resolve these NS servers did not produce any substantive A or MX records.
- **Containing A or MX records:** 95% (5,358 domains) had available A records (pointing to IPv4 addresses) or MX records (indicating mail servers). These domains are likely in normal or partially normal use, making them difficult to classify as parked.

4.1.4 Conclusion

By my definition, domains with no DNS information and only NS records are considered parked domains when registered for more than three months. This represents 5% of domains in a sample of 10,000 domains that are initially considered parked domains.

4.2 Parked Domain Statistics

4.2.1 Overall statistics

Based on the study's definition of "broadly parked domains": Among the 5,656 domains in the "registered for more than three months" category, 298 domains met these criteria: 169 with no records at all and 129 with only NS records without further resolution. This represents 2.98% of the entire sample of 10,000 domains.

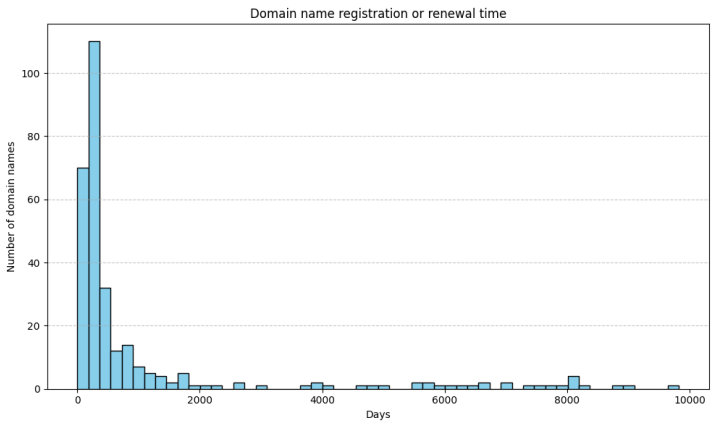

Figure 4.1: Parking time distribution of 298 parked domains.

In total, 298 domains were identified as parked. Each domain in this set meets the following criteria:

1. It has a valid WHOIS record (i.e., the domain is registered).

2. It lacks any functional DNS entries beyond possible NS placeholders (i.e., no resolvable `A`, `MX`, or `CNAME` records).

3. The registration duration (*days*), calculated from the creation or last update time to the current time, is at least three months.

4.2.2 Box Plot Outlier Threshold.

A box plot analysis of the *days* distribution revealed:

- First quartile (Q1): 186 days
- Third quartile (Q3): 721.75 days
- Interquartile range (IQR) = Q3 - Q1 = 535.75 days
- Outlier cut-off (upper bound) = Q3 + 1.5 × IQR ≈ 1,525.375 days

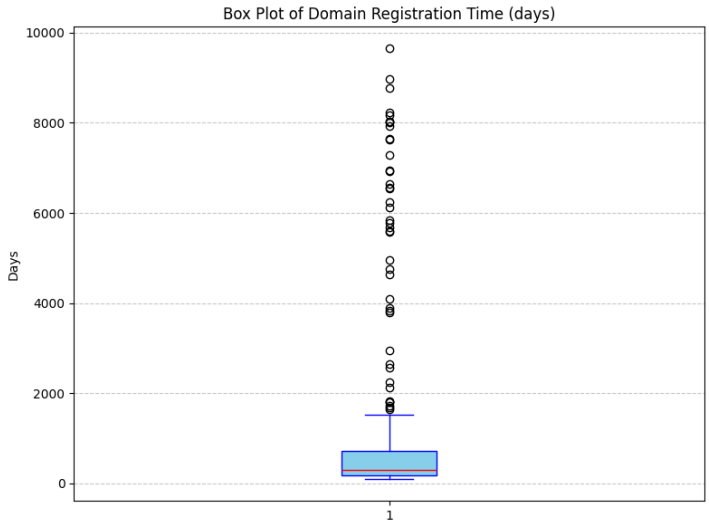

Figure 4.2: Parking time distribution of 298 parked domains in box plot.

Any domain with $days > 1,525$ is considered an outlier. According to this criterion, 42 of the 298 parked domains exceed 1,525 days, while 256 remain below that boundary.

4.2.3 Non-Outlier Parked Domains ($days \leq 1,525$)

These 256 domains have registration durations ranging roughly from 90 days up to just under 4.2 years. Key observations:

Registration Recency and Distribution

- **Median:** 307 days. At least half of these domains have been registered for under one year.
- **25th Percentile (Q1):** 186 days, indicating one-fourth of these domains have been registered for about half a year or less.
- **75th Percentile (Q3):** 721.75 days, corresponding to nearly two years of continuous holding.

Most non-outlier domains appear relatively new, suggesting a vibrant market in which domain names are frequently and recently registered, possibly with short- to medium-term speculative horizons.

Potential Motivations

These domains lack resolvable `A` or `MX` records (beyond potential `NS` placeholders). Owners may:

- **Speculate on Future Value:** Expecting certain keywords, short numeric combos, or brandable phrases to become valuable.
- **Minimize Overheads:** Avoiding hosting costs and restricting expenses to registration renewals.
- **Gauge Demand:** Some domains may be silently listed on aftermarket platforms; owners keep them offline unless a sale materializes.

Domain Name Characteristics and TLD Patterns

- **Short Alphanumeric Variants**: Many feature concise strings (e.g., `72g.com`, `5i01.com`), often favored in certain markets (e.g., numeric domains in Chinese-speaking regions).
- **Keyword-Focused Names**: Examples like `autocarindonesia.com`, `allmarketingtrends.com` signal owners banking on brand or industry relevance.
- **Mixed TLD Distribution**: While `.com` recurs frequently, ccTLDs such as `.in`, `.de`, `.tn`, and `.us` also appear. In contrast to the outlier group, many non-outlier ccTLD domains appear to be newer registrations, reflecting more recent speculative opportunities.

Summary (Non-Outliers)

Most of these parked domains are under four years old, highlighting a wave of recent registrations for speculative or protective reasons. The low-maintenance approach (no `A` or `MX` records) reduces holding costs, aligning with short- to mid-term investment objectives.

4.2.4 Outlier Parked Domains ($\boldsymbol{days} > 1,525$)

A smaller subset of 42 domains surpasses the 1,525 day threshold, indicating registration ages of over 4.2 years. Some are closer to 20+ years of holding time (e.g., `aids.ru` at 9,653 days).

Extended Registration Durations

- **Range:** From just over 4 years to nearly 26 years.
- **Continuous Renewals:** Paying renewal fees for such a prolonged period points to strong expectations of future payoffs or an owner's reluctance to relinquish potentially valuable real estate.

Statistical Characteristic

- **Right-Skewed Distribution**: Just 14% of parked domains crossing 4+ years drastically inflate the mean (1,072.52 days) and standard deviation (1,983.97).
- **Long-Horizon Speculation**: Owners of these domains exhibit greater patience and resource commitment than typical short-term speculators, demonstrating that domain parking can become a multi-decade phenomenon.
- **International Diversity**: The strong presence of `.cn` and `.ru` outliers emphasizes that long-term domain hoarding is not exclusive to `.com` but is geographically widespread.

TLD Distribution Skew

Notably, a large share of these outliers are ccTLD domains such as `.ru` and `.cn`, with examples including:

- `aids.ru` (9,653 days)
- `abkhazia.ru` (8,974 days)
- `agnitum.ru` (8,174 days)
- `7b.cn` (7,936 days)

- `alimama.cn` (8,023 days)
- `agri.cn` (8,029 days)

This indicates that long-term holding behavior is especially prominent for specific ccTLD markets—likely due to earlier domain adoption in those regions, local brand significance, or historically cheaper renewal rates. Some `.ro`, `.cl`, and `.ac.cn` domains also appear, further illustrating a geographically diverse but older inventory within the outlier set.

Potential Reasons for Decade-Long Parking

- **Historic/High-Value Names**: Short numeric or keyword-rich `.cn` and `.ru` domains can carry elevated resale prices, especially in niche markets.
- **Legacy Usage**: Certain outliers may have hosted sites in the early 2000s or late 1990s; after abandonment, owners maintained registration (removing DNS entries but continuing to renew).
- **Strategic Brand/Keyword Control**: Entities might have acquired these domains to block rivals or to preserve future brand expansions in specific TLDs without immediate development.

Summary (Outliers)

These long-held domain names reflect a strategy of enduring investment or holding indefinitely, favoring mainly ccTLDs such as `.cn` and `.ru`. They form a small but influential subset, highlighting how domain name parking can last for decades, driven perhaps by brand protection or high-value keywords specific to certain languages and regions.

4.2.5 Different Parking Strategies and ccTLD Concentration Patterns

1. **Two Different Strategies for Parking Domains**: Non-outlier domains (shorter holds) show recent, active speculation, while outliers (longer holds) point to more entrenched, high-expectation investments.
2. **Outliers are Mainly Concentrated in ccTLDs**: The outlier group features a markedly higher proportion of `.cn`, `.ru`, and other country-code domains. Early adopters in these TLDs may have recognized strategic keywords or numeric combinations and maintained them for decades.

4.2.6 Conclusion.

In summary, the 298 parked domains show a clear polarization: the majority are relatively new registrations (shorter than 4 years old) reflecting a dynamic speculative and quick turnaround mentality, contrasted with a smaller but significant number of older domains (usually ccTLDs) that can be held for more than 20 years. Overall, this stratified holding pattern of parked domains reflects the existence of two typical types of parked domains, long term and short term.

4.3 Further Analysis of Whois Information for Parked Domains

This section provides a comprehensive analysis of 298 parking domains along a number of dimensions:

- **TLD Distribution:** Proportions of ccTLD vs. gTLD domains, followed by complete tables of each category.
- **WHOIS Privacy and Security:** Key attributes such as WHOIS privacy, EPP lock, and DNSSEC adoption.
- **Registrar Overview:** A full list of all registrars appearing in the dataset, with counts of domains per registrar.

4.3.1 TLD Distribution and Proportions

Based on the dataset, we have identified:

- **ccTLD domains:** 70
- **gTLD domains:** 228

Hence, about 24% of the parked domains are ccTLDs, while 76% fall under gTLDs. The sections below detail each category.

ccTLD Listing and Analysis

Table 3.4 shows that, after excluding samples with only a single instance, the average parked durations for Russia (.ru), China (.cn), and Romania (.ro) are 5,704 days, 5,584 days, and 4,661 days, respectively. These figures indicate that domains in these regions have been held for significantly longer periods compared to other ccTLDs. Such extended parking durations may be attributed to several factors:

ccTLD	Sum Count	Total Days	Avg. Days/Domain
cl (Chile)	1	6,535	6,535
ly (Libya)	1	5,832	5,832
ru (Russia)	10	57,040	5,704
cn (China)	20	111,682	5,584
ro (Romania)	5	23,306	4,661
th (Thailand)	1	3,897	3,897
in (India)	4	6,653	1,663
de (Germany)	2	2,713	1,356
tn (Tunisia)	2	2,440	1,220
uz (Uzbekistan)	2	1,168	584
ca (Canada)	2	784	392
fi (Finland)	1	337	337
cc (Cocos Islands)	2	490	245
it (Italy)	5	1,074	214
tv (Tuvalu)	3	643	214
me (Montenegro)	1	209	209
pl (Poland)	2	399	199
us (United States)	1	195	195
fr (France)	3	525	175
se (Sweden)	1	162	162
co (Colombia)	1	105	105

Table 4.4: Complete List of Parked ccTLD Domains:

gTLD Listing and Analysis

The data in Table 3.5 confirms that the `.com` TLD is the most prevalent, with 191 domains and an average parking time of 432 days. This is significantly shorter than the multi-year averages for several ccTLDs (e.g., 5,704 days for `.ru`, 5,584 days for `.cn`, and 4,661 days for `.ro`). These differences do suggest that `.com` domains are flipped or reused more frequently, likely due to their global popularity and higher liquidity in the aftermarket.

gTLD	Sum Count	Total Days	Avg. Days/Domain
com	191	82,658	432
net	27	8,069	298
info	4	670	167
org	6	2,025	337

Table 4.5: Complete List of Parked gTLD Domains

4.3.2 WHOIS Privacy and Security Attributes

By thoroughly examining the WHOIS information for these parked domains, I found that these domains exhibit distinctive characteristics in terms of privacy protection and security. Table 3.6 shows the common characteristics of these parked domains.

Attribute	Sum Count
WHOIS privacy (hidden registrant info)	226
EPP lock (protected against transfer)	196
DNSSEC not registered	226

Table 4.6: Number of domains with this attribute

Analysis of WHOIS Attributes.

- **High rate of WHOIS privacy (226 domains)**: Over 76% of domain owners opt to hide their registration details, a common practice among domain investors and corporate brand-protection strategies.
- **EPP lock used by 196 domains**: About two-thirds of the sample have EPP lock enabled, indicating awareness of unauthorized transfer risks. However, roughly one-third remain vulnerable to hijacking or accidental transfer.
- **DNSSEC adopted**: 226 domains do not have DNSSEC. Since these domains are parked rather than actively hosting critical or transactional services, owners may deprioritize DNSSEC setup.

4.3.3 Registrar Distribution and Analysis

By counting the registrants in the whois information in the parked domains. I found some similarities between the registrants or registrars of these parked domains.Figure 3.7 illustrates the registrant distribution information for parked domains.

Registrant Identification Analysis

Most of these "registrants" are domain registrars. This group also includes some companies that provide enterprise brand protection, bulk domain management, or advanced domain hosting services. Their common feature is that they are either qualified for domain registration or have agency rights with the respective top-level domain registries (Registry), enabling them to register and manage domain names for end users (individuals, businesses, organizations).They can be roughly divided into the following categories:

Registrar	Sum Count
GoDaddy.com, LLC	47
Alibaba Cloud Computing (Beijing) Co., Ltd.	19
eName Technology Co.,Ltd.	11
Network Solutions, LLC	10
RU-CENTER-RU	10
NAMECHEAP INC	9
MarkMonitor, Inc.	7
TUCOWS, INC.	7
NameSilo, LLC	7
CSC CORPORATE DOMAINS, INC.	6
GMO Internet, Inc.	6
Alibaba Cloud Computing Ltd. (Wanwang)	6
Name.com, Inc.	5
ENOM, INC.	5
PDR Ltd. d/b/a PublicDomainRegistry.com	5
DYNADOT LLC	5
CSL Computer Service Langenbach GmbH d/b/a joker.com	4
Alibaba Cloud Computing Ltd. d/b/a HiChina (www.net.cn)	4
Xin Net Technology Corporation	4
Hefei Juming Network Technology Co., Ltd.	4
DNSPod, Inc.	3
Amazon Registrar, Inc.	3
Xiamen Ename Technology Co., Ltd.	3
ICI - Registrar	3
1API GmbH	3
Registrars with one or two parked domain names	85

Table 4.7: Number of times a domain name from each domain name registrar appears in the sample

Major International Generic Registrars

- For example: GoDaddy.com, LLC; Namecheap; Network Solutions; Tucows; eNom; IONOS; MarkMonitor; CSC Corporate Domains; Gandi; Porkbun; Name.com; Dynadot; NameSilo; etc.

Large number of Chinese registrars

- For example: Alibaba Cloud Computing Co., Ltd. (Wangwang); eName (Xiamen Ename Technology); Chengdu West Dimension Digital; Shanghai Meicheng Technology Information Development Co., Ltd.; Beijing Xinwang Digital Information Technology Co., Ltd.; Foshan Yidong Network Co., Ltd.; Hainan Meijie Technology Co., Ltd.; Hefei Juming Network Technology; Hongkong Kouming International Limited; Beijing Guoxu Network Technology Co., Ltd.; Xiamen Sanwu Internet Information Co., Ltd.; etc.

Regional/Country-Specific Registrars

- For example: RU-CENTER (Russia); THNIC (Thailand); JPRS (Japan); NIC Chile (Chile); ERNET India (India); Aftermarket.pl (Poland); OVH (France); Register S.p.a. (Italy); acens Technologies (Spain); WebCC (Malaysia/Singapore); nazwa.pl (Poland); etc.

Enterprise-Level Domain and Intellectual Property Protection Service Providers

- For example: Corporate Domains; CSC Corporate Domains; MarkMonitor; NOM-IQ dba Com Laude; etc.

Small or Relatively New Registrars

- For example: SAV.COM; Allworldnames.com LLC; 1800-website, LLC; DNC Holdings, Inc.; etc.

Geographical Distribution of Registrants

Based on the company registration information or headquarters location, these registrars can be roughly categorized into the following major regions:

- **North America**
  - Typical examples: GoDaddy (USA); Namecheap (USA); MarkMonitor (USA); Network Solutions (USA); Tucows (Canada); eNom (USA); Porkbun (USA); Name.com (USA); NameSilo (USA); Dynadot (USA); IONOS (despite its German background, it also has significant operations in the USA).

- **Europe**
  - For example: Gandi (France); IONOS (Germany); OVH (France); Register S.p.a. (Italy); acens Technologies (Spain); Metaregistrar BV (Netherlands); Realtime Register B.V. (Netherlands); NOM-IQ Ltd dba Com Laude (UK).
- **Asia**
  - Mainland China: Alibaba Cloud (Wangwang); eName (Ename Technology); Chengdu West Dimension Digital; Beijing Xinwang; Shanghai Meicheng; Foshan Yidong; Hainan Meijie; Beijing Guoxu; Xiamen Sanwu Internet; Hefei Juming; Hongkong Kouming.
  - Japan: GMO Internet; JPRS; etc.
  - Singapore: Gname.com Pte. Ltd.
- **Other Regions**
  - Russia: RU-CENTER.
  - Thailand: THNIC.
  - India: ERNET India; BigRock Solutions.
  - Poland: Aftermarket.pl; nazwa.pl.
  - Chile: NIC Chile.
  - Libya: Libyan Spider Network.
  - Israel: Domain The Net Technologies.

Numerical Characteristics

Based on the sample data, it is evident that GoDaddy is far ahead of the others, while the remaining registrars are more evenly distributed.

- For example, there are many registrars in the list that only show 1, 2, or 3 domains.
- This indicates a long-tail effect in the sample: a few large registrars account for the majority of domains, while many small or specialized registrars have only a few domains each.

Summary

Through statistical analysis of the registrants in the whois information of parked domain names, I found that most of the registrants are actually domain name registrars with domain name registration or agency rights, which also include enterprises providing corporate brand protection, large-scale domain name management or advanced hosting services. These registrars can be broadly categorised into large international general-purpose registrars, regional or national registrars, enterprise-level domain name and IP protection service providers, and some smaller or emerging registrars. Geographically, registrars are located in North America, Europe, Asia and other regions. The sample data shows that a few large registrars (e.g., GoDaddy) account for the majority of registrations, while the rest of the registrars show a long-tailed distribution with only a small number of domain names registered.

4.3.4 Conclusion

By analysing the TLD categories, WHOIS attributes and registrar distribution of parked domain names. I discovered:

- **TLD Distribution:**
  - In terms of total percentage, the percentage of parked domains in gTLDs is larger than that in ccTLDs, but in terms of parking time, it can be found that the parking time of parked domains in ccTLDs is larger than that in gTLDs.
- **WHOIS Protection and Security:**
  - The whois information of parked domains reveals that most of the domains hide the actual information of the registrant. And most of the domains have EPP lock enabled. On the contrary, most of the parked domains are in DNSSEC not registered status.
- **Registrar Landscape:**
  - Most of the parked domain names are registered by domain name registrars with registration credentials.

Chapter 5

Conclusion

Based on OpenIntel's publicly available DNS measurements and combining WHOIS information with DNS record queries, this study defines and identifies Parked Domains from a broader perspective. After analysing a random sample of 10,000 domains, the main conclusions and findings are as follows.

5.1 Recognition methods

The program first obtains key information such as registration length and registration status of domain names through batch WHOIS queries. The program filters out domain names that have been registered for more than three months, and then conducts multiple rounds of DNS record queries for these domain names. If there is no available A-record or mail resolution pointing to the domain name and it does not contain normal CNAME, it will be finally decided as a parked domain name. If there are no available A-records or email resolutions pointing to the domain name and the domain name does not contain a normal CNAME, then the domain name is ultimately determined to be a parked domain name. I used large-scale domain name resolution data provided by OpenIntel as a sample, combined with local automated scripts, de-duplication and random sampling strategies to obtain a representative subset for analysis. This identification method captures more domains that have been vacant for a long time but are still registered than a narrow solution that focuses only on whether or not the domain jumps to an advertisement page.

5.2 Results

About 57% of the domain names in the sample have been registered for more than three months, of which about 3% were eventually identified as parked domain names with no actual content or valid resolution records. This category of parked domains includes both

domains that were registered between a few months and a few years ago. It also includes parked domains that have been held for more than 10 or even 20 years. Among them, gTLDs such as `.com` have the largest number, while some ccTLDs such as `.ru` and `.cn` have been parked for a longer period of time on average.

More than 70% of these parked domain names choose to hide registrant information, and a significant proportion of them have EPP lock enabled, while most of them still do not have DNSSEC configured. This reflects that the holders attach great importance to the protection of their own privacy and the prevention of unauthorised transfers, while investing relatively little in security features such as DNSSEC. Parked domain names are mostly registered by international domain name registrars and regional registrars with large-scale registration qualifications, and the leading registrars occupy a larger share, exhibiting a clear long-tail distribution.

5.3 Summary

In summary, based on large-scale DNS data, this paper explores the feasibility of using WHOIS and DNS multilayer queries to identify broadly-defined parked domain names, and obtains observable quantitative results. Although more in-depth analyses, such as port-level detection are still lacking. This study has demonstrated that a significant proportion of parked domain names can be efficiently identified by starting from the registration length and DNS record characteristics.

5.4 Future Work

As a next step, I can extend the detection to common ports such as HTTP/HTTPS, SMTP, SSH, and so on. It can also be integrated with web content collection when needed. Thus, it can more accurately determine whether the domain name is really in a state of no business operation.

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